Cylindrical coordinate system

Dr. Sampath Kumar

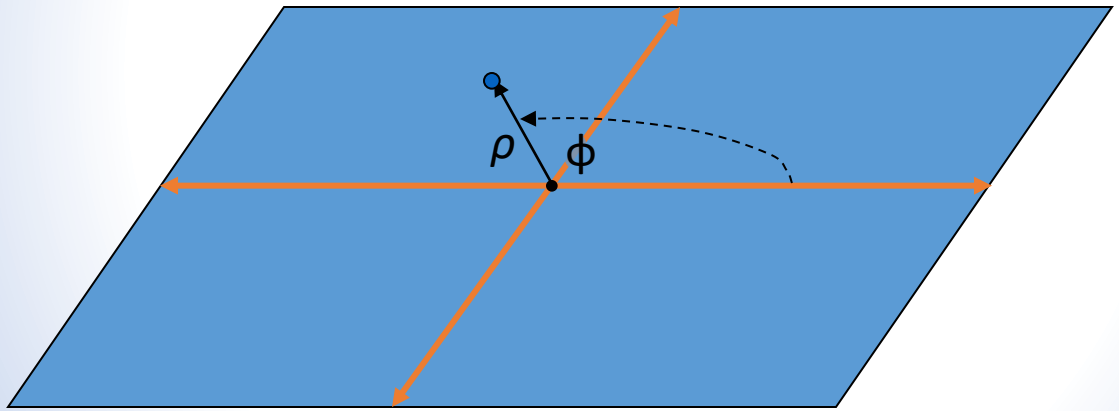
Associate Professor

Department of ECE

MIT, Manipal

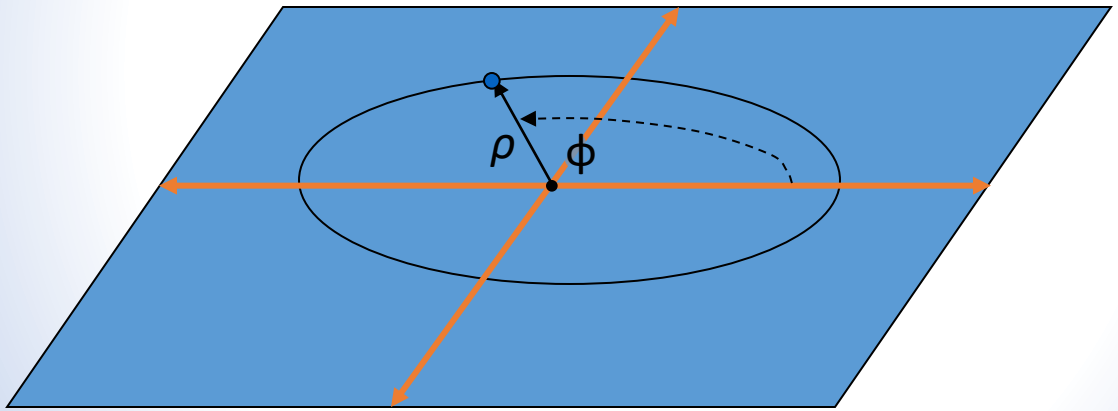
Representing 3D points in Cylindrical Coordinates.

Recall polar representations in the plane



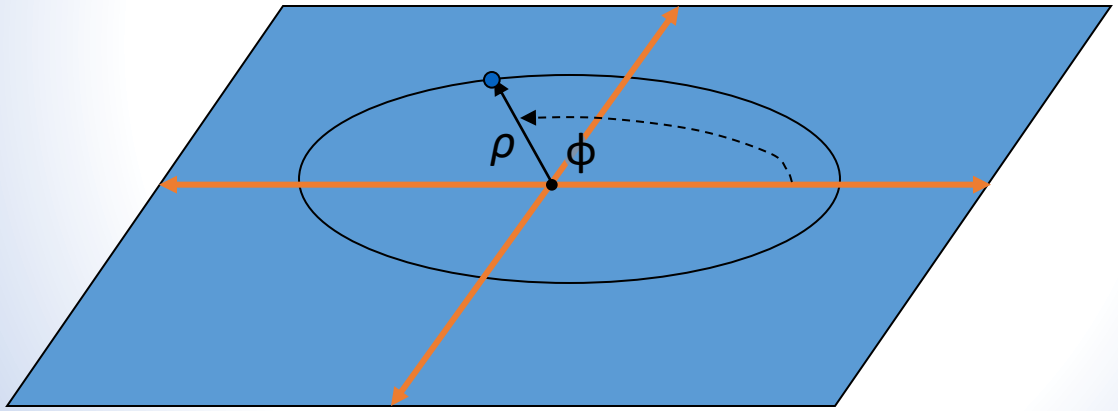
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Cylindrical coordinates just adds a *z*-coordinate to the polar coordinates (ρ, ϕ).



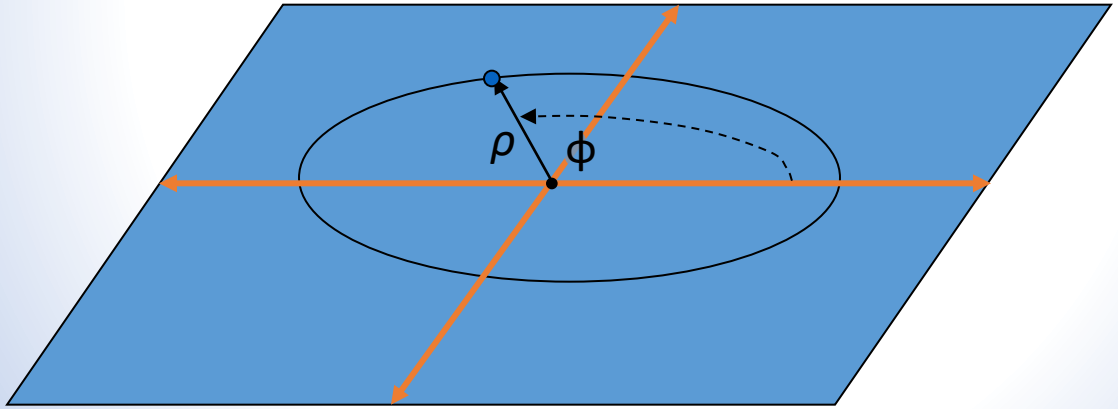
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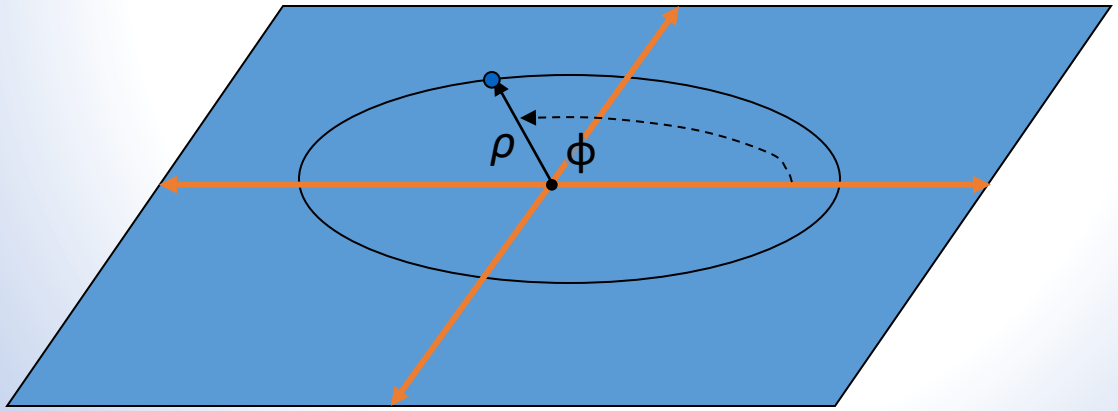
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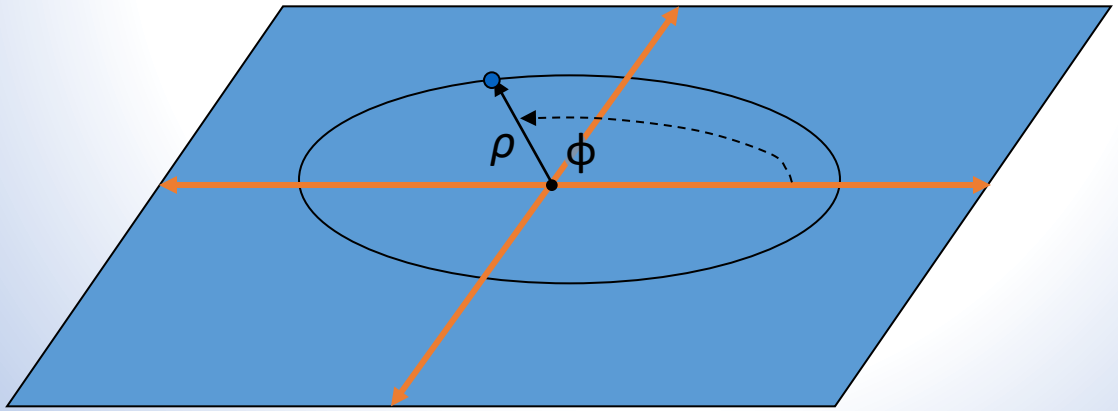
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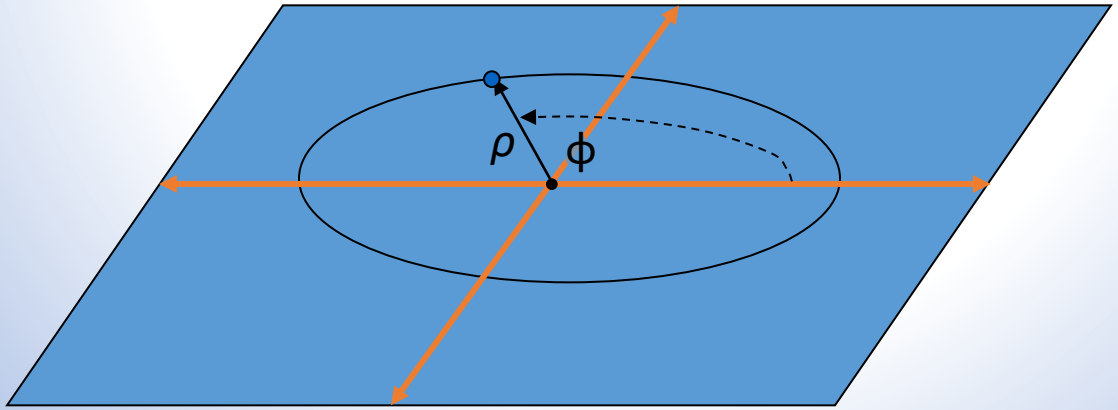
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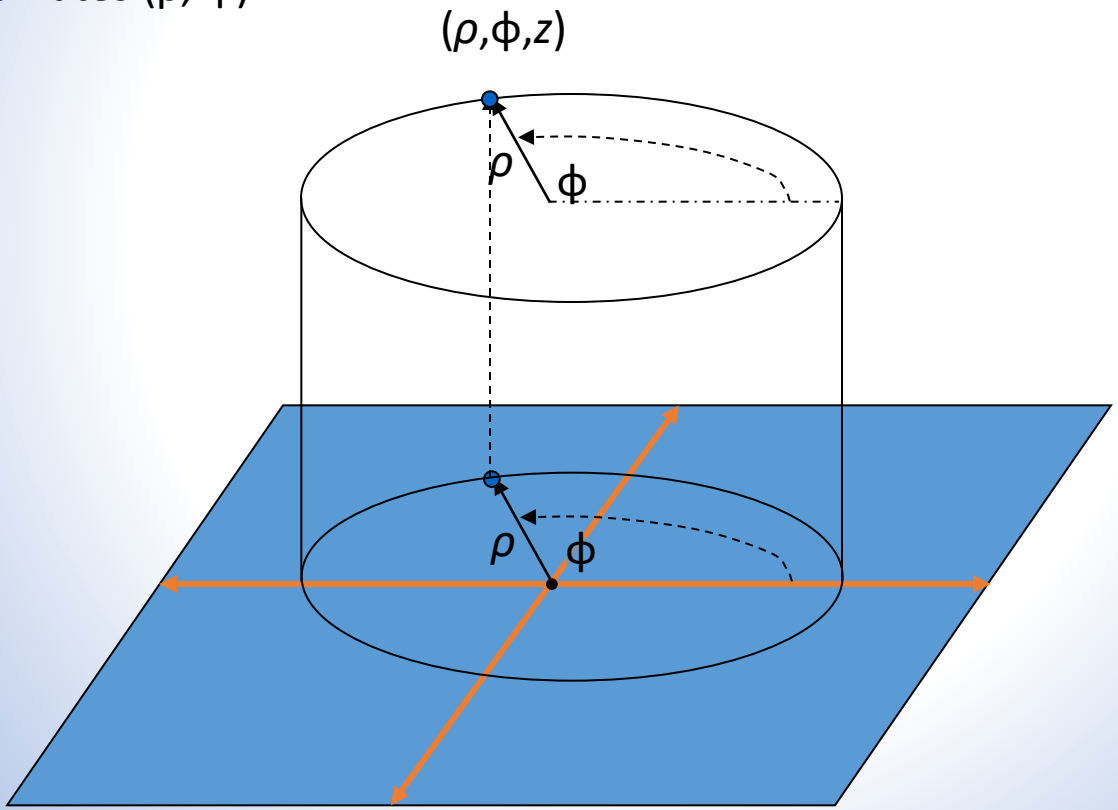
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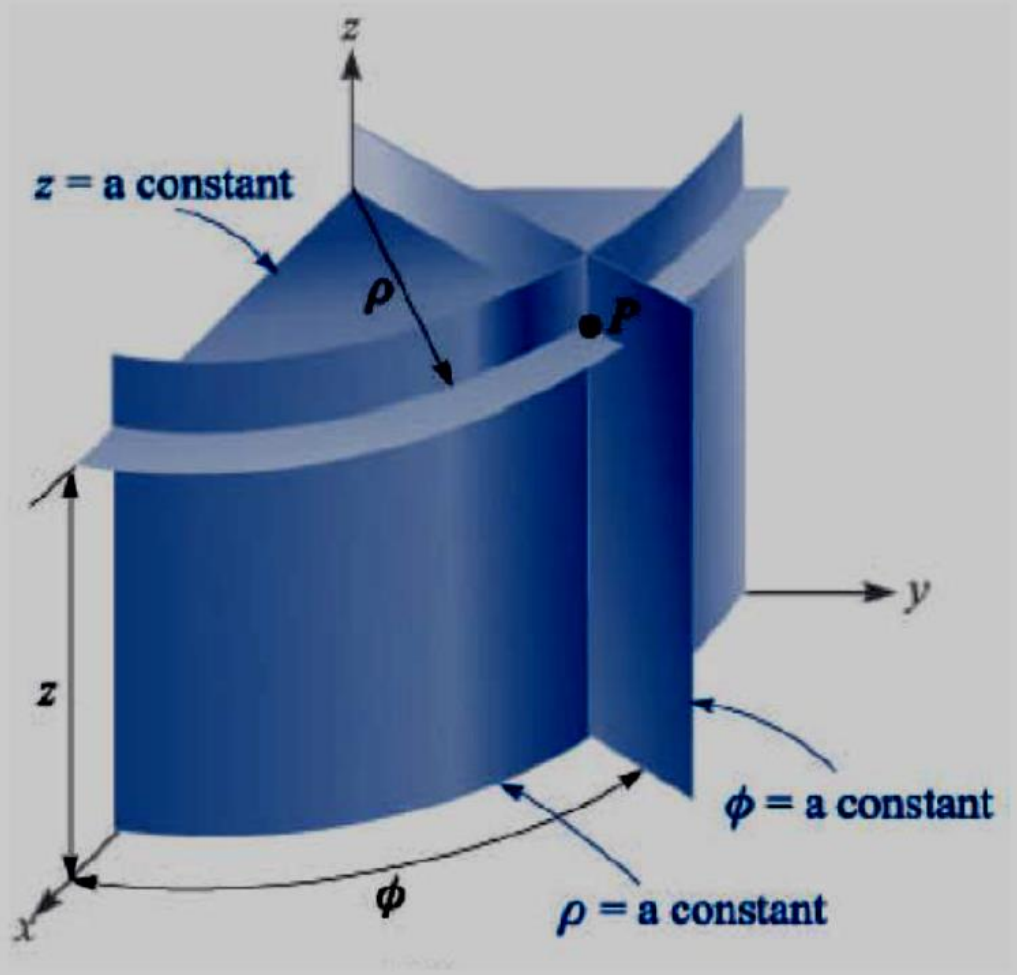


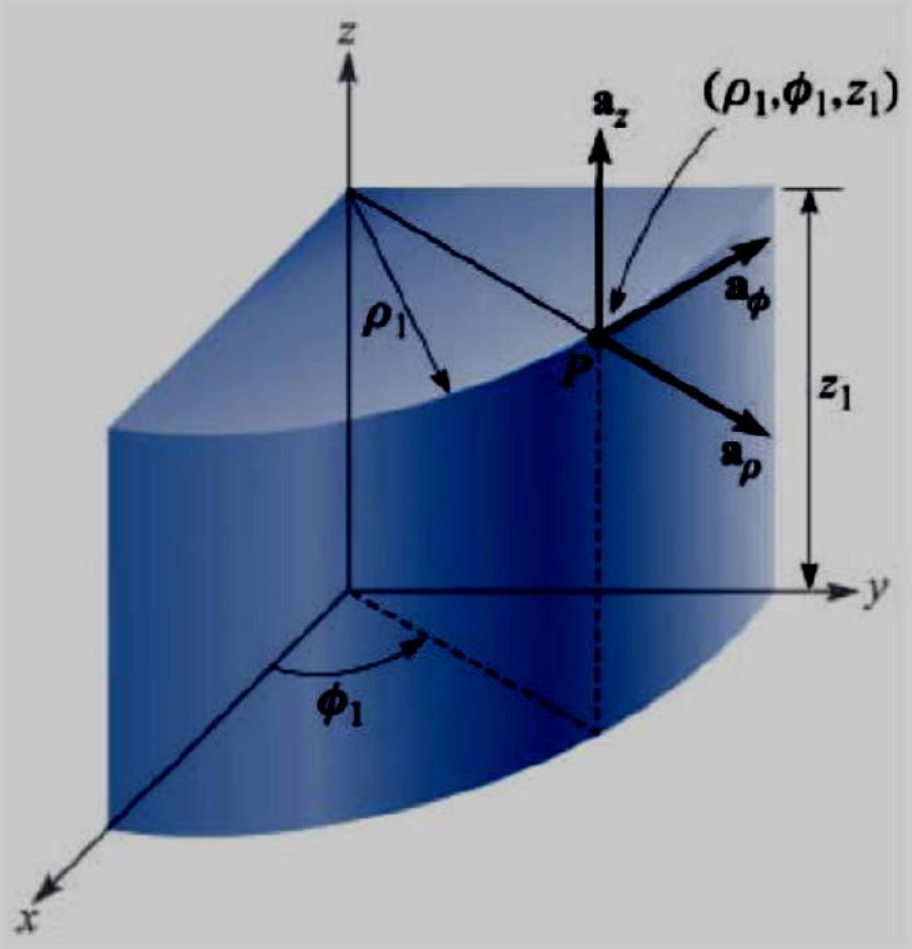
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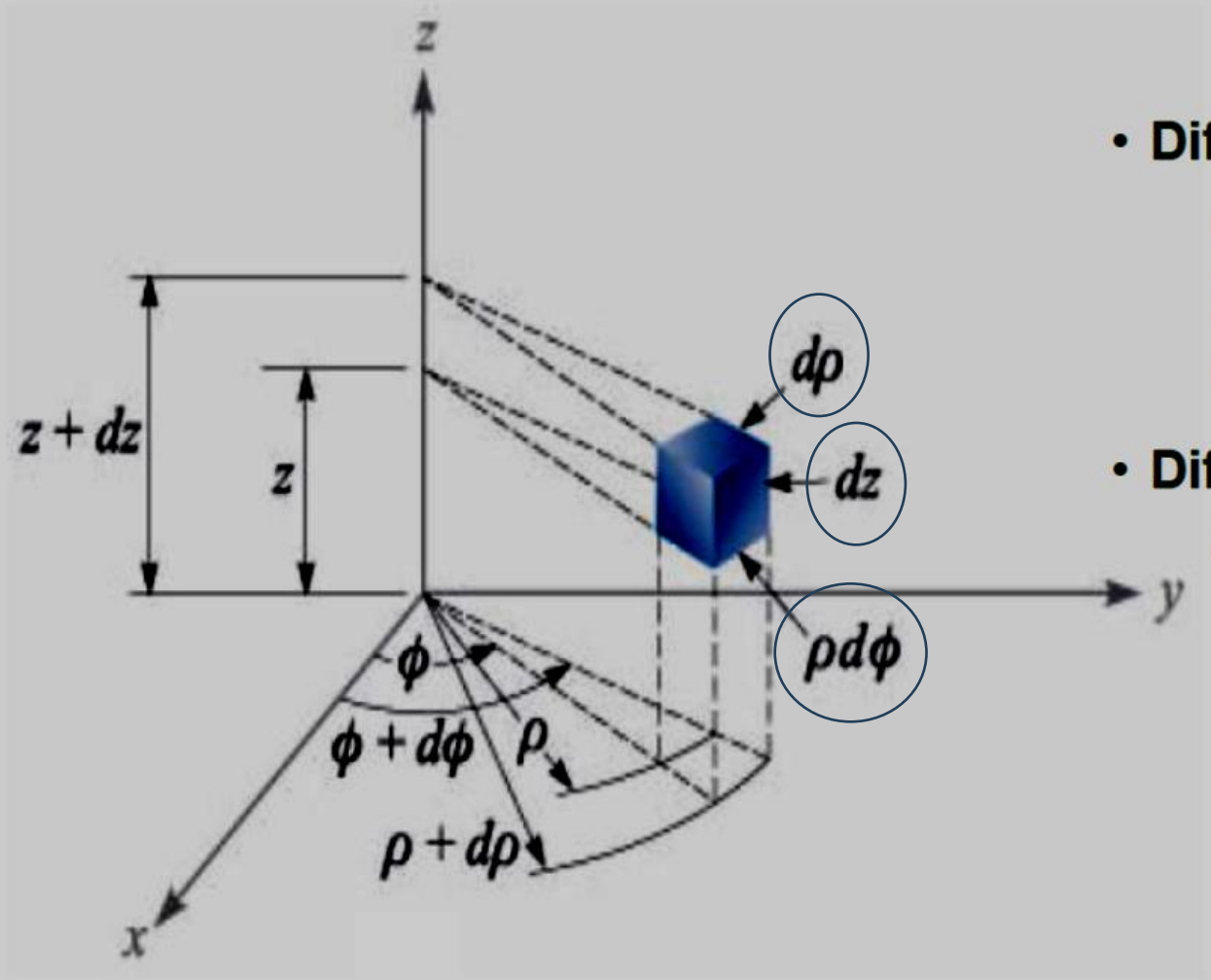


The Cylindrical Coordinate System





The Cylindrical Coordinate System



- **Differential surface units:**

  $d\rho \cdot dz$

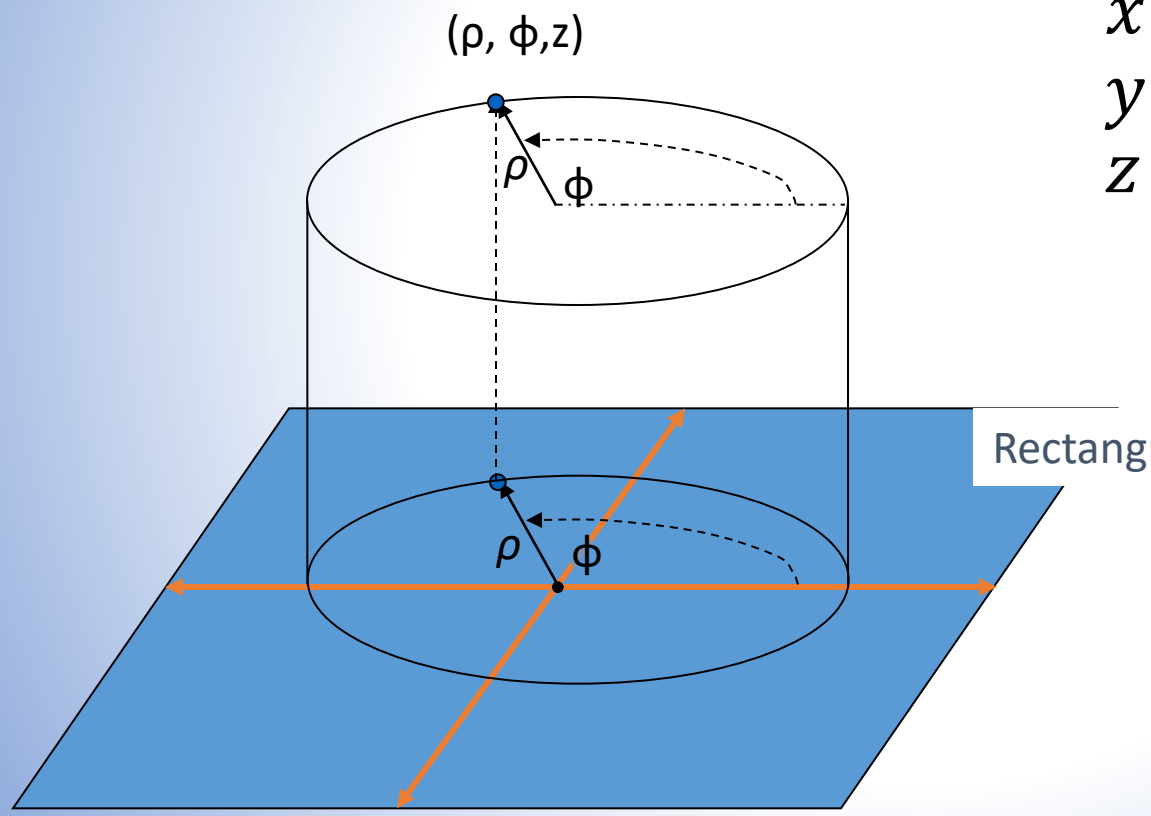
  $\rho d\phi \cdot dz$

  $d\rho \cdot \rho d\phi$

- **Differential volume unit :**

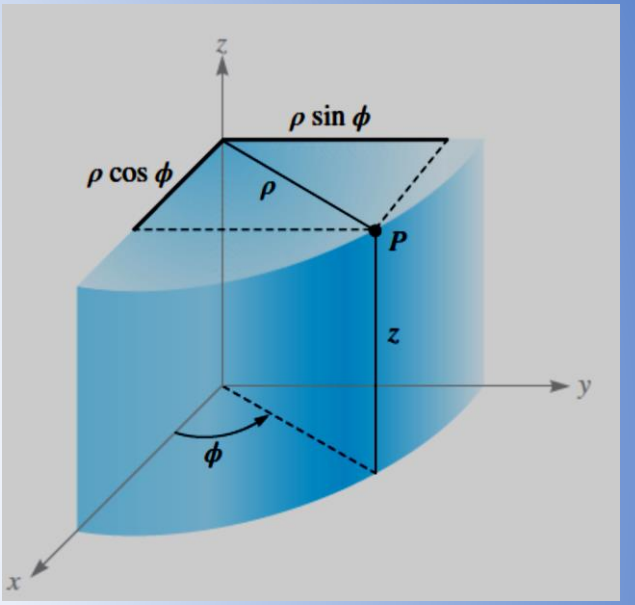
  $d\rho \cdot \rho d\phi \cdot dz$

Converting between rectangular and Cylindrical Coordinates



Cylindrical to rectangular

$$x = \rho\cos(\phi)$$
$$y = \rho\sin(\phi)$$
$$z = z$$



Rectangular to Cylindrical

$$\rho^2 = x^2 + y^2$$
$$\tan(\phi) = \frac{y}{x}$$
$$z = z$$

$$\rho = \sqrt{x^2 + y^2} \quad (\rho \geq 0)$$
$$\phi = \tan^{-1}\frac{y}{x}$$
$$z = z$$

Convert point P (3, 4, 5) in Cartesian coordinate to cylindrical?

$$r = \sqrt{3^2 + 4^2} = 5$$

$$\phi = \tan^{-1} 4/3 = 0.927\, rad$$

$$z = 5$$

D1.5. (*a*) Give the rectangular coordinates of the point $C(\rho = 4.4, \phi = -115°, z = 2)$. (*b*) Give the cylindrical coordinates of the point $D(x = -3.1, y = 2.6, z = -3)$. (*c*) Specify the distance from C to D.

a) $x = \rho\cos\phi = 4.4\cos(-115°) = -1.86$

$y = \rho\sin\phi = -3.99$

$z = 2$

$C(-1.86, -3.99, 2)$

$D = (x = -3.1, y = 2.6, z = -3)$

$\rho = \sqrt{x^2+y^2} = 4.06$

$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(0.670)$ $\quad -39.98°$ or $140°$

$\bar{R}_{CD} = \bar{\rho}_D - \bar{\rho}_C$

$= -1.24\bar{a}_x + 6.59\bar{a}_y - 5\bar{a}_z$

$|\bar{R}_{CD}| = 8.36$

- The previous example works fine if scalar functions are transformed
- Following method is used for vector function transformation

$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$$

$$\mathbf{A} = A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$$

$$A_\rho = \mathbf{A}\cdot\mathbf{a}_\rho \qquad \text{and} \qquad A_\phi = \mathbf{A}\cdot\mathbf{a}_\phi$$

$$A_\rho = (A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z)\cdot\mathbf{a}_\rho = A_x\mathbf{a}_x\cdot\mathbf{a}_\rho + A_y\mathbf{a}_y\cdot\mathbf{a}_\rho$$

$$A_\phi = (A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z)\cdot\mathbf{a}_\phi = A_x\mathbf{a}_x\cdot\mathbf{a}_\phi + A_y\mathbf{a}_y\cdot\mathbf{a}_\phi$$

$$A_z = (A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z)\cdot\mathbf{a}_z = A_z\mathbf{a}_z\cdot\mathbf{a}_z = A_z$$

Table 1.1 Dot products of unit vectors in cylindrical and rectangular coordinate systems

	$\mathbf{a}_\rho$	$\mathbf{a}_\phi$	$\mathbf{a}_z$
$\mathbf{a}_x\cdot$	$\cos\phi$	$-\sin\phi$	0
$\mathbf{a}_y\cdot$	$\sin\phi$	$\cos\phi$	0
$\mathbf{a}_z\cdot$	0	0	1

D1.6. Transform to cylindrical coordinates: (*a*) $\mathbf{F} = 10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z$ at point $P(10, -8, 6)$; (*b*) $\mathbf{G} = (2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y$ at point $Q(\rho, \phi, z)$. (*c*) Give the rectangular components of the vector $\mathbf{H} = 20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z$ at $P(x = 5, y = 2, z = -1)$.

$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$$

$$\mathbf{A} = A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$$

$$A_\rho = \mathbf{A} \cdot \mathbf{a}_\rho \quad \text{and} \quad A_\phi = \mathbf{A} \cdot \mathbf{a}_\phi$$

Table 1.1 Dot products of unit vectors in cylindrical and rectangular coordinate systems

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$$A_\rho = (A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z) \cdot \mathbf{a}_\rho = A_x\mathbf{a}_x \cdot \mathbf{a}_\rho + A_y\mathbf{a}_y \cdot \mathbf{a}_\rho = 10\,\hat{a}_x \cdot \hat{a}_\rho - 8\hat{a}_y \cdot \hat{a}_\rho$$

$$A_\rho = 10\cos\phi\,\hat{a}_\rho - 8\sin\phi\,\hat{a}_\rho$$

$$A_\phi = (A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z) \cdot \mathbf{a}_\phi = A_x\mathbf{a}_x \cdot \mathbf{a}_\phi + A_y\mathbf{a}_y \cdot \mathbf{a}_\phi = 10\hat{a}_x \cdot \hat{a}_\phi - 8\hat{a}_y \cdot \hat{a}_\phi$$

$$A_\phi = -10\sin\phi\,\hat{a}_\phi - 8\cos\phi\,\hat{a}_\phi$$

$$P(10,-8,6) \Rightarrow \phi = \tan^{-1}\left(\frac{-8}{10}\right) = -38.66^{\circ}$$

$$A_\rho = 10\cos\phi\,\hat{a}_\rho - 8\sin\phi\,\hat{a}_\rho$$

$$A_\rho = 12.81\,\hat{a}_\rho$$

$$A_\phi = -10\sin\phi\,\hat{a}_\phi - 8\cos\phi\,\hat{a}_\phi$$

$$A_\phi = 0$$

$$a_2 = a_2 = 6\hat{a}_2$$

$$\vec{F} = 12.81\hat{a}_\rho + 6\hat{a}_2$$

D1.6. Transform to cylindrical coordinates: (*a*) $\mathbf{F} = 10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z$ at point $P(10, -8, 6)$; (*b*) $\mathbf{G} = (2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y$ at point $Q(\rho, \phi, z)$. (*c*) Give the rectangular components of the vector $\mathbf{H} = 20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z$ at $P(x = 5, y = 2, z = -1)$.

Ans. (*a*) $12.81\mathbf{a}_\rho + 6\mathbf{a}_z$; (*b*) $(2\rho\cos^2\phi - \rho\sin^2\phi + 5\rho\sin\phi\cos\phi)\mathbf{a}_\rho + (4\rho\cos^2\phi - \rho\sin^2\phi - 3\rho\sin\phi\cos\phi)\mathbf{a}_\phi$; (*c*) $H_x = 22.3, H_y = -1.857, H_z = 3$

Note: Part (a) is already solved in previous slide. Try part (b) and (c)


Thank
you